

Ryland Gross

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EDUCATION

Imperial College London

MSc in Pure Mathematics

London, United Kingdom

Expected June 2027

Harvard University

A.B. *cum laude* in Mathematics

Citation in Russian Language

Cambridge, MA

May 2026

RESEARCH INTERESTS

Commutative algebra; combinatorics; formal verification of mathematics; automated theorem proving; AI for math.

RESEARCH EXPERIENCE

Autoformalization of Commutative Algebra in Lean 4

2025–present

Independent Research (GitHub)

Developing a Lean 4 / Mathlib formalization of results from Atiyah and Macdonald's *Introduction to Commutative Algebra*. Current work includes formalization of results in ring theory, ideals, modules, and dimension theory. The formalization is also being used to construct a benchmark for evaluating AI-assisted theorem proving in commutative algebra.

Harvard Mathematics REU

May–August 2025

Research Investigator, supervised by Dr. Philip Matchett Wood

Cambridge, MA

Conducted research in combinatorial matrix theory, focusing on polytope varieties, Birkhoff polytope classes, and near-derangement matrices. Developed a depth-first search algorithm for maximizing trace over this matrix class and obtained new lower bounds for exclusive near-derangement total support matrices trace.

ArcGIS Spatial Analysis, Kentucky Geologic Survey

May–August 2023

Research Intern

Lexington and Hazard, KY

Collaborated with the Kentucky Geologic Survey and the Housing Development Alliance on GIS-based methods for identifying viable housing sites for Eastern Kentucky flood-displacement victims. Identified 15 candidate building sites; 2 homes were subsequently constructed.

SOFTWARE AND PRESENTATIONS

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- **LeanTeX** (GitHub; documentation) — A L^AT_EX package and Python command-line tool for embedding Lean 4 proof snippets and Infoview output into mathematical documents.
 - *GIS-Based Housing Site Identification for Eastern Kentucky Flood Victims*. Appalachian Research Conference, 2023.

TEACHING EXPERIENCE

Harvard University

Course Assistant

Cambridge, MA

September 2023–May 2026

Served as course assistant for three undergraduate courses:

- Math 22b: Linear Algebra and Vector Analysis
- Math 130: Classical Geometry (Euclidean and Non-Euclidean)
- Math 157: Mathematics in the World (Head CA)

Taught students topics such as linear algebra, multivariable calculus, Euclidean and non-Euclidean geometry, introductory probability, elementary complex analysis, cryptography, Python programming, and proofs.

OUTREACH AND SERVICE

Prison Mathematics Project

Remote / Cambridge, MA

Mentor

October 2025–April 2026

Provided one-on-one mathematical mentorship to an incarcerated individual, covering topics from elementary algebra and calculus through number theory, proof-writing, and abstract algebra.

Community Web Development, Eastern Kentucky

2025–present

Volunteer

Designed and maintained websites for local churches, nonprofits, and small businesses in Eastern Kentucky. Recent projects include a memorial foundation website for the Team Colton DIPG Foundation supporting pediatric brain cancer awareness.

GRADUATE AND ADVANCED COURSEWORK

Math 221: Graduate Commutative Algebra

Math 243: Graduate Evolutionary Dynamics

Math 161: Formal Methods in Mathematics (Lean 4 / Mathlib)

Math 114: Real Analysis II

Math 124: Number Theory

Math 131: Topology and the Fundamental Group

Math 113: Complex Analysis

Math 136: Differential Geometry

Math 122: Abstract Algebra I and II

Statistics 111: Introduction to Statistical Inference

Statistics 110: Probability Theory

TECHNICAL SKILLS

Proof Assistants: Lean 4 / Mathlib

Programming Languages: Lean 4, Python, Rust, R

Tools: Git, L^AT_EX, ArcGIS

CITIZENSHIP AND LANGUAGES

Citizenship: United States

Languages: English (native), Russian (reading)